Correlation

Correlation is a statistical tool which measures the degree of relationship between any two variable characteristics.

Karl Pearson's coefficient of correlation:

For frequency distribution the formula is

$$r = \frac{\frac{\sum xy}{n} - \overline{x}\overline{y}}{\sigma_x \sigma_y}$$ **, when the series is individual series**

For grouped data,

$$r = \frac{\frac{\sum fxy}{\sum f} - \overline{x}\overline{y}}{\sigma_x \sigma_y}$$ **where** $\sigma_x = \sqrt{\frac{1}{n}\sum x^2 - \bar{x}^2}; \quad \sigma_y = \sqrt{\frac{1}{n}\sum y^2 - \bar{y}^2}$

Correlation coefficient (r) lies between -1 and 1.

1. **Calculate the correlation coefficient for the following heights(in inches) of fathers X and their sons Y.**

X	**65**	**66**	**67**	**67**	**68**	**69**	**70**	**72**
Y	**67**	**68**	**65**	**68**	**72**	**72**	**69**	**71**

X	Y	XY	X^2	Y^2
65	67	4355	4225	4489
66	68	4488	4356	4624
67	65	4355	4489	4225
67	68	4556	4489	4624
68	72	4896	4624	5184
69	72	4968	4761	5184
70	69	4830	4900	4761
72	71	5112	5184	5041
$\sum X = 544$	$\sum Y = 552$	$\sum XY = 37560$	$\sum X^2 = 37028$	$\sum Y^2 = 38132$

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$$\overline{X} = \frac{\sum X}{n} = \frac{544}{8} = 68$$

$$\overline{Y} = \frac{\sum Y}{n} = \frac{552}{8} = 69$$

$$\overline{X}\,\overline{Y} = 4962$$

$$\sigma_x = \sqrt{\frac{1}{n}\sum x^2 - \bar{x}^2} = \sqrt{\frac{37028}{8} - 4624} = 2.121$$

$$\sigma_y = \sqrt{\frac{1}{n}\sum y^2 - \bar{y}^2} = \sqrt{\frac{38132}{8} - 4761} = 2.345$$

$$r = \frac{\frac{\sum xy}{n} - \overline{x}\overline{y}}{\sigma_x \sigma_y} = \frac{\frac{37560}{8} - 4962}{2.121 X\ 2.345} = 0.6030$$

2. Find the correlation coefficient between X and Y.

X	**1**	**2**	**3**	**4**	**5**
Y	**2**	**5**	**3**	**8**	**7**

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